



FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

IT0049 (WEB SYSTEM TECHNOLOGIES)

TECHNICAL FORMATIVE ASSESSMENT

2

FROM ARRAYS TO A REAL DATABASE

Student Name / Group Name:	ISAIAH EZEKIEL P. VICENCIO
Section:	BSIT BA - TW32
Professor:	VON ERICK MAGBITANG



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PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements. [PO: C]

COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- Apply advanced web development principles in creating, debugging, validating, and securing database-backed web applications. [CLO: 2]

INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- Configure CodeIgniter's database connection settings.
- Create a MySQL database and tables using a provided schema.
- Build CodeIgniter Models for the Customer Accounts and User Accounts pages.
- Retrieve records with Query Builder and display them in a view.

BACKGROUND INFORMATION

The POS application from the previous activity stored its data in static PHP arrays, which reset every time the server restarts. This activity replaces that temporary storage with a real MySQL database. CodeIgniter's Model class wraps a database table and gives you Query Builder methods — such as `findAll()` — for retrieving records without writing raw SQL.

A good rule for this activity: once a page's data comes from a Model, its controller and view code should look almost identical to the static-array version. Only where the data comes from changes.



GRADING SYSTEM / RUBRIC

CRITERIA	Description
Functionality & Requirements Completeness (40 points max)	All required pages and features are present and work as specified in the activity instructions.
Code Structure & Organization (25 points max)	Controllers, models, and views are separated correctly and follow the framework's conventions; code is readable and sensibly named.
Database Design & Query Usage (20 points max)	The database and tables match the provided schema, records are retrieved through a Model using Query Builder (not raw SQL), and the results display correctly in the view.
Documentation & Submission Quality (15 points max)	The GitHub repository is organized and complete, the README explains how to set up and run the project, and the hosted link is live and matches the submitted code.



LABORATORY ACTIVITY

You will extend the POS application from TFA1 by replacing its static arrays with a real MySQL database. Use the schema below.

1. Configure your CodeIgniter .env database settings to connect to a local MySQL database.
2. Create the database and the customers and users tables using the schema below:

```
CREATE TABLE customers (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  full_name VARCHAR(100) NOT NULL,  
  email VARCHAR(100) NOT NULL,  
  phone VARCHAR(20),  
  created_at DATETIME NOT NULL  
);  
  
CREATE TABLE users (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  username VARCHAR(50) NOT NULL UNIQUE,  
  full_name VARCHAR(100) NOT NULL,  
  created_at DATETIME NOT NULL  
);
```

3. Populate each table with at least 5 sample records.
4. Create a CustomerModel and a UserModel.
5. Update your Customer Accounts and User Accounts controllers to retrieve records through their Models instead of static arrays.
6. Confirm the views display the retrieved database records the same way they displayed the static array data in TFA1.

Submission Requirements:

- A link to a GitHub repository containing your raw project files, including your database export.
- A link to your hosted, working version of the application.



POS Management / Home

CodeIgniter 4 · MySQL

Point-of-sale workspace

Browse customer contact information and staff account details from one focused management interface.

[View customers](#) [View staff users](#)

SYSTEM OVERVIEW

A focused directory for essential POS accounts.

The application demonstrates a clean CodeIgniter MVC workflow backed by the required MySQL customer and user tables.

Framework	CodeIgniter 4
Data source	MySQL database
Architecture	Model → Controller → View

QUICK ACCESS

Account directories

Open a database-backed directory to review its current records.

01 Customer Accounts
Names, email addresses, phone numbers, and creation dates.

→

02 User Accounts
Usernames, staff names, and account creation dates.

→

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Figure 1. Home Screen

POS Management / Customer Accounts

CodeIgniter 4 · MySQL

Customer Accounts

5
TOTAL RECORDS

Contact details and account metadata for customers stored in the POS database.

DATABASE DIRECTORY

Customer records

[customers table](#)

ID	FULL NAME	EMAIL	PHONE	CREATED AT
#1	Angela Reyes	angela.reyes@example.com	+63 917 245 8103	Sep 1, 2026 · 9:15 AM
#2	Marco Villanueva	marco.villanueva@example.com	+63 918 632 4751	Sep 2, 2026 · 10:30 AM
#3	Sofia Mendoza	sofia.mendoza@example.com	+63 905 381 9264	Sep 3, 2026 · 11:45 AM
#4	Daniel Lim	daniel.lim@example.com	+63 922 714 5386	Sep 4, 2026 · 1:20 PM
#5	Bianca Santos	bianca.santos@example.com	+63 916 849 2075	Sep 5, 2026 · 3:10 PM

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Figure 2. Customer Accounts



POS System

Management console

WORKSPACE

Home

Customer Accounts

User Accounts

About

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POS Management / User Accounts

CodeIgniter 4 - MySQL

STAFF DIRECTORY

User Accounts

5
TOTAL RECORDS

Identity and account metadata for staff users stored in the POS database.

DATABASE DIRECTORY

Staff user records

users table

ID	USERNAME	FULL NAME	CREATED AT
#1	acruz	Andrea Cruz	Sep 1, 2026 · 8:00 AM
#2	jnavarro	Joshua Navarro	Sep 2, 2026 · 8:30 AM
#3	mpascual	Mikaela Pascual	Sep 3, 2026 · 9:00 AM
#4	rgonzales	Rafael Gonzales	Sep 4, 2026 · 9:30 AM
#5	tgarcia	Trisha Garcia	Sep 5, 2026 · 10:00 AM

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Figure 3. User Accounts

POS System

Management console

WORKSPACE

Home

Customer Accounts

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POS Management / About

CodeIgniter 4 - MySQL

ABOUT THE PROJECT

CodeIgniter 4 POS System

This learning project demonstrates how database records move through CodeIgniter models, controllers, and views in a simple MVC application.

ASSESSMENT CONTEXT

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The application was created for IT0049 - Web System Technologies. It uses dedicated controllers to prepare page data and reusable views to present a consistent, responsive interface.

APPLICATION FLOW

Clear MVC responsibilities

Database → Model → Controller → View → HTML

DATA LAYER

Database-backed records

Customer and user records come from the required MySQL tables through CodeIgniter models. The controllers pass those records to the views, where `foreach` loops build the account tables.

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Figure 4. About Page



```
# Local development settings. This file is intentionally ignored by Git.
CI_ENVIRONMENT = development

app.baseURL = 'http://localhost:8080/'
app.indexPage = ''

# Local MySQL connection for TFA2. Change these values if your setup differs.
database.default.hostname = 127.0.0.1
database.default.database = pos_db
database.default.username = root
database.default.password =
database.default.DBDriver = MySQLi
database.default.DBPrefix =
database.default.port = 3306
```

Figure 4. CodeIgniter Database Configuration

```
[mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| pos_db |
| sys |
+-----+
5 rows in set (0.031 sec)

[mysql> USE pos_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
[mysql> SHOW TABLES;
+-----+
| Tables_in_pos_db |
+-----+
| customers |
| users |
+-----+
2 rows in set (0.003 sec)
```

Figure 5. MySQL `pos_db` database containing the required `customers` and `users` tables



```
[mysql> DESCRIBE customers;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(100)	NO		NULL	
phone	varchar(20)	YES		NULL	
created_at	datetime	NO		NULL	

```
5 rows in set (0.007 sec)
```

Figure 6. Structure of the **customers** MySQL table

```
[mysql> SELECT * FROM customers;
```

id	full_name	email	phone	created_at
1	Angela Reyes	angela.reyes@example.com	+63 917 245 8103	2026-09-01 09:15:00
2	Marco Villanueva	marco.villanueva@example.com	+63 918 632 4751	2026-09-02 10:30:00
3	Sofia Mendoza	sofia.mendoza@example.com	+63 905 381 9264	2026-09-03 11:45:00
4	Daniel Lim	daniel.lim@example.com	+63 922 714 5386	2026-09-04 13:20:00
5	Bianca Santos	bianca.santos@example.com	+63 916 849 2075	2026-09-05 15:10:00

```
5 rows in set (0.002 sec)
```

Figure 7. Five sample records stored in the **customers** table

```
[mysql> DESCRIBE users;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
username	varchar(50)	NO	UNI	NULL	
full_name	varchar(100)	NO		NULL	
created_at	datetime	NO		NULL	

```
4 rows in set (0.003 sec)
```

Figure 8. Structure of the **users** MySQL table



```
[mysql> SELECT * FROM users;
```

id	username	full_name	created_at
1	acruz	Andrea Cruz	2026-09-01 08:00:00
2	jnavarro	Joshua Navarro	2026-09-02 08:30:00
3	mpascual	Mikaela Pascual	2026-09-03 09:00:00
4	rgonzales	Rafael Gonzales	2026-09-04 09:30:00
5	tgarcia	Trisha Garcia	2026-09-05 10:00:00

```
5 rows in set (0.001 sec)
```

Figure 9. Five sample records stored in the **users** table

REFERENCES

- Manfield, A. (2021). Pro CodeIgniter: Learn how to create professional web-applications with PHP.
- CodeIgniter User Guide. Models.
<https://codeigniter4.github.io/userguide/models/model.html>
- CodeIgniter User Guide. Query Builder.
https://codeigniter4.github.io/userguide/database/query_builder.html